

# Intelligent Syllabus Mapping and Performance Analysis System Using NLP and Neural Networks

1<sup>st</sup> Vivin Lobo  
Dept. of CSE  
Sahyadri College of Engineering  
and Management  
Mangaluru, India  
vivinlobo.dev@gmail.com

2<sup>nd</sup> Shreya Bhaskar  
Dept. of CSE  
Sahyadri College of Engineering  
and Management  
Mangaluru, India  
shreyabhaskar1991@gmail.com

3<sup>rd</sup> Sourabha S  
Dept. of CSE  
Sahyadri College of Engineering  
and Management  
Mangaluru, India  
sourabhas2603@gmail.com

4<sup>th</sup> Royden Suvarna  
Dept. of CSE  
Sahyadri College of Engineering  
and Management  
Mangaluru, India  
royden0714@gmail.com

5<sup>th</sup> Vidya V V  
Dept. of CSE  
Sahyadri College of Engineering  
and Management  
Mangaluru, India  
vidyav.cs@sahyadri.edu.in

**Abstract**—The rapid evolution of educational standards requires efficient methods for aligning course content with learning objectives and evaluating student performance. This paper presents an approach for extracting and mapping syllabus content to learning outcomes using Natural Language Processing (NLP) techniques and advanced neural networks. The system integrates text preprocessing, keyword extraction, and web scraping to map course material with predefined learning objectives. A neural-network based performance analysis module further assesses student progress based on mapped content. The proposed system ensures comprehensive coverage of educational standards while minimizing gaps and redundancies. Using NLP and neural networks, the system streamlines curriculum management, enhances personalized learning, and improves educational planning and assessment.

**Keywords**—Automated Content Extraction, Data-Driven Approach, Educational Planning, Instructional Strategies, Natural Language Processing (NLP), Neural Networks, Performance Analysis, Personalized Learning, Student Performance Data

## I. INTRODUCTION

The rapidly evolving landscape of education presents students with the challenge of managing vast amount of study material, especially during exam preparation. With limited time and an overwhelming volume of content to review, students often struggle to organize and prioritize their study resources effectively. The traditional approach of manually gathering and structuring study materials can be time-consuming and stressful, leading to poor time management and unstable academic performance. Moreover, students have diverse learning preferences—some grasp concepts better through visual aids like videos and diagrams, while others prefer textual content. Conventional study methods, such as reviewing scattered notes or conducting unstructured internet searches, often fail to accommodate these varying learning styles. As a result, students

may find it difficult to focus on essential topics, leading to inefficiencies in their study process.

### A. The Need for Intelligent Syllabus Mapping Solutions

Traditional study methods have limitations, so we introduce a system designed to address common challenges students face while studying. It utilizes advanced technologies like Natural Language Processing and Neural Networks to enhance learning efficiency. The primary objective of this system is to simplify studying by organizing syllabus content into an interactive dashboard. It converts PDF syllabus into a structured and personalized format, allowing students to track their progress and focus on key topics. By automating the process of organizing study materials, this system helps save time, making learning more efficient and stress-free.

### B. Functionality: From PDF to Personalized Dashboard

The core functionality of the system begins with analyzing a syllabus provided in PDF format. Using natural language processing techniques and pattern recognition, the system extracts key concepts, topics, and learning objectives from the document. These topics are then categorized and structured into a personalized dashboard, offering students an organized view of their study material. To enhance learning, the system integrates automated content retrieval methods to gather relevant study materials. It uses web scraping to collect online articles and reference materials, while video analysis techniques help identify high-quality educational videos based on relevance and ratings. This ensures that students have access to diverse and credible resources tailored to their syllabus. In addition, students can assess their understanding using quizzes based on the extracted topics. The system tracks study patterns and quiz performance, allowing a neural network model to

predict the future academic performance of a student. This prediction is based on factors such as study duration, quiz scores, and class test results, helping students identify areas that need improvement and optimize their learning strategy.

### C. Automation, Personalization, and Data-Driven Learning

The proposed system revolutionizes study practices by automating the tedious process of gathering and organizing study materials while personalizing the learning experience. Instead of students manually compiling notes and searching for resources, the system extracts key topics, curates relevant study materials, and structures them into an optimized study path. The personalized dashboard adapts to individual learning preferences, offering study resources in multiple articles, videos, and diagrams, ensuring an engaging and efficient learning experience.

Beyond automation, the system leverages data-driven insights to predict student performance. By tracking quiz scores and study time on the platform, the neural network model forecasts future performance trends. This helps students assess their progress and stay informed about their learning trajectory. By providing a structured and efficient study process, the system reduces exam stress and enhances overall academic preparation.

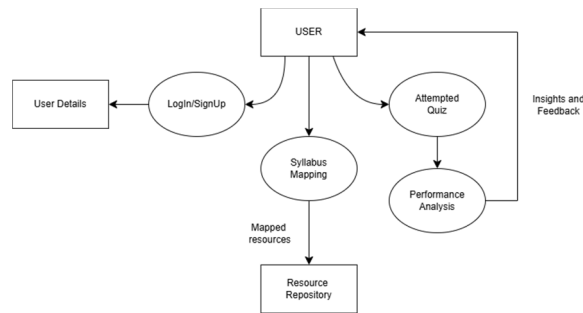


Fig. 1: Overview of System Flow and Data Mapping

Figure 1 outlines the system’s flow, from user login to accessing syllabus-mapped resources, attempting quizzes, and receiving performance insights. The system analyzes quiz scores and study time to generate feedback, helping users track their progress.

## II. RELATED WORKS

Krishna et al. (2020) presented a study that focuses on utilizing Natural Language Processing (NLP) for syllabus mapping. Their research achieved high accuracy in aligning course content with learning outcomes, demonstrating how automation can enhance curriculum design. By saving time and resources, NLP enables educators to focus more on teaching quality and student engagement rather than manual syllabus alignment. The study also highlights the role of NLP in improving clarity and coherence in course structures, ensuring effective delivery of learning outcomes [1].

Khan and Ghosh (2020) reviewed 140 studies on student performance prediction, identifying critical performance predictors and emphasizing the timing of predictions. Their

findings revealed that predictive models perform well during course tenure, but early predictions—made before the course begins—remain underdeveloped. This gap highlights a crucial area for further research to ensure timely interventions for struggling students, ultimately improving overall academic performance [2].

Abubakari and Suprpto (2020) explored the use of neural networks for predicting student performance and achieved over 96% accuracy. Their work demonstrates the potential of deep learning models in educational data mining, proving their reliability for analysing academic outcomes. However, while the study prioritized high prediction accuracy, its scope was limited to performance tracking. Our proposed system builds upon this by integrating predictive capabilities with personalized resource recommendations, offering a comprehensive solution for learning optimization [3].

Nathoo (2022) investigated NLP algorithms for generating adaptive learning materials. The study demonstrated how NLP can analyse educational content, extract key concepts, and recommend tailored study resources. This advancement enables intelligent platforms to provide adaptive learning materials based on syllabus content. Incorporating these findings into our system ensures that students receive personalized and relevant study materials, fostering a more effective learning experience [4].

Abodayeh et al. (2024) explored the effectiveness of web scraping using Python’s BeautifulSoup library for data extraction. Their study showed how web scraping can efficiently collect syllabus data from online sources. By integrating web scraping techniques into our system, we enable the automated extraction of syllabus content from diverse platforms. This further enhances the system’s ability to map curricula and recommend relevant learning resources, improving overall performance in educational settings [5].

The paper “Early Predicting of Students’ Performance in Higher Education” examines machine learning and educational data mining for forecasting academic outcomes. It emphasizes early prediction to identify at-risk students using features like admission scores, gender, and early-course grades. Methods include clustering, classification, and t-SNE, with models like Random Forest, SVM, and XGBoost showing improved accuracy when combining admission scores with early grades. The study reviews deep learning, fairness-aware algorithms, and ensemble methods, highlighting the role of computational techniques in enhancing educational strategies and recommending the integration of non-academic factors for future research [6].

The paper “Prediction of Student’s Academic Performance through Data Mining Approach” (Hussain et al., 2024) explores Educational Data Mining (EDM) for predicting academic performance using machine learning. Techniques like artificial neural networks (ANN) and deep learning achieved up to 88.6% accuracy by analyzing test scores, attendance, and assignments. Integrating socioeconomic factors improved insights, though challenges like data limitations, model complexity, and privacy concerns remain. The study advocates

for incorporating psychological and social variables and larger datasets to enhance predictions, emphasizing the role of data-driven approaches in identifying at-risk students and improving educational outcomes [7].

The paper “An Analysis of Application of Natural Language Processing in Enhancing Education” (Raj and Paliwal, 2021) examines NLP’s role in modernizing education through automated assessment, language evaluation, and curriculum enhancement. Applications like automated essay scoring, error detection, and intelligent tutoring systems improve learner engagement and personalized learning. While NLP-driven platforms enhance writing and comprehension skills, challenges remain in contextual understanding and integration into diverse educational settings. The study underscores future research directions to refine NLP tools for interactive learning and error detection, highlighting its impact on improving educational access and quality [8].

The paper “Predicting Students’ Academic Performance in Educational Data Mining Based on Deep Learning Using TensorFlow” (Abubakaria et al., 2020) explores machine learning (ML) and educational data mining (EDM) for academic performance prediction. It reviews methodologies like clustering, classification, and feature selection, alongside advanced techniques such as hybrid regression, ensemble methods, and deep learning. Key factors influencing predictions include academic records, entrance scores, demographics, and first-year performance. The study emphasizes integrating diverse academic and behavioral features with ML algorithms to enhance prediction accuracy and support educational planning [9].

The paper “A Survey of the Usages of Deep Learning for Natural Language Processing” (Otter et al., 2020) provides an overview of deep learning advancements in NLP. It examines how models like RNNs, LSTMs, and transformer-based architectures (BERT, GPT) have improved tasks such as machine translation, sentiment analysis, and speech recognition. Key concepts include word embeddings, attention mechanisms, and transfer learning. The study also discusses challenges like data requirements, computational demands, and model interpretability, highlighting future research directions for improving NLP systems [10].

The paper “Web Scraping for Data Analytics: A BeautifulSoup Implementation” (Abodayeh et al., IEEE) explores web scraping techniques using Python’s BeautifulSoup. It highlights how web scraping extracts structured data from unstructured content, aiding data analytics where traditional datasets are unavailable. The study details HTML/XML parsing with BeautifulSoup, addressing challenges like dynamic content, legal considerations, and ethical concerns. Practical applications include market analysis, sentiment analysis, and competitive intelligence, with a case study demonstrating real-world implementation. The paper underscores web scraping’s role in data-driven decision-making and its integration with machine learning and analytics workflows [11].

### III. METHODOLOGY

The proposed methodology for the Intelligent Syllabus Mapping and Performance Analysis System focuses on creating an automated and efficient platform that addresses the common challenges students face in managing their academic progress. Traditional syllabus mapping methods are often time-consuming and require manual effort to align topics with study resources. This system leverages NLP, neural networks, and external APIs to automate the syllabus mapping process, making it more accurate and tailored to students’ individual learning needs. The methodology consists of five major components:

#### A. Syllabus Upload and Processing

- Syllabus Upload: Students upload their syllabus in PDF format through a frontend interface.
- NLP Processing: The backend utilizes Python scripts and NLTK to extract topics and keywords relevant to the syllabus content.

#### B. Resource Recommendation

- Video Recommendations: The system integrates with the YouTube V3 API to fetch videos based on extracted topics.
- Article Retrieval: Web scraping or Google search APIs are used to gather related articles for each topic.
- Database Storage: All recommended resources are categorized and stored for future reference.

#### C. Performance Analysis

- Neural Network Integration: The system utilizes a Multi-Layer Perceptron (MLP) neural network to analyze student assessment data, track academic progress, and predict performance trends.
- Personalized Insights: The platform presents performance insights through graphs and charts, enabling students to analyze their strengths and weaknesses effectively.

#### D. System Architecture

- Backend Coordination: A Flask-based backend manages interactions between the frontend, NLP engine, APIs, and the database.
- Database Management: A MongoDB database stores uploaded syllabi, mapped resources, and student performance data.
- Frontend Interface: Built with React and Next.js, the frontend provides a seamless user experience for accessing study materials and performance analytics.

#### E. Workflow Overview

- Syllabus Processing: Students upload their syllabus, triggering NLP-based topic extraction and content mapping.
- Resource Fetching: The system retrieves relevant study resources and stores them in the database.
- Performance Updates: Assessment data is dynamically updated, offering real-time insights to students.

## IV. PROPOSED ALGORITHM

### A. Algorithm 1: PDF Upload and Text Processing

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#### Algorithm 1 Handling PDF Upload and Preprocessing

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**Require:** Syllabus PDF file uploaded by the user

**Ensure:** Tokenized and cleaned text

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1: Step 1: File Handling
2: if file.extension  $\neq$  ".pdf" then
3:   Reject file and return an error message
4: end if
5: if file.size > MAX_SIZE then
6:   Reject file and return an error message
7: end if
8: filePath  $\leftarrow$  SaveFileToTempLocation(file)
9: Step 2: Text Extraction and Cleaning
10: rawText  $\leftarrow$  ExtractTextFromPDF(filePath)
11: cleanedText  $\leftarrow$  RemoveSpecialCharacters(rawText)
12: tokens  $\leftarrow$  Tokenize(cleanedText)
13: tokens  $\leftarrow$  RemoveStopwords(tokens)
14: tokens  $\leftarrow$  ApplyStemmingOrLemmatization(tokens)
15: return tokens

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The process of PDF upload and text extraction ensures that syllabus data is structured into a format suitable for further processing. By eliminating irrelevant characters and stopwords, the algorithm improves the accuracy of subsequent NLP-based operations, such as topic extraction and mapping.

### B. Algorithm 2: Concept Identification and Content Mapping

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#### Algorithm 2 Extracting Key Concepts and Mapping Resources

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**Require:** Tokenized syllabus text

**Ensure:** Mapped educational resources

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1: Step 1: Key Concept Extraction
2: keyConcepts  $\leftarrow$  ExtractTopicsFromSyllabus(tokens)
3: Step 2: Content Mapping
4: for each concept in keyConcepts do
5:   videos  $\leftarrow$  FetchYouTubeLinks(concept)
6:   articles  $\leftarrow$  ExtractArticleLinks(concept)
7:   StoreMappedResources(concept, videos, articles)
8: end for
9: return MappedResources

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The identification of key concepts through POS tagging, NER, and topic modeling allows for efficient extraction of syllabus topics. These topics are then mapped to relevant online resources, enhancing accessibility to curated educational materials for students.

### C. Algorithm 3: Recommendation and Performance Analysis

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#### Algorithm 3 Personalized Recommendations and Performance Prediction

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**Require:** Content mapping, user profile, and interaction data

**Ensure:** Final recommendations and performance insights

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1: Step 1: Personalized Resource Recommendation
2: Initialize recommendations dictionary
3: for each topic in mapping do
4:   recs[topic]  $\leftarrow$  Personalize(mapping[topic], profile)
5: end for
6: Step 2: Performance Analysis and Prediction
7: modelInput  $\leftarrow$  PrepareModelInput(interactionData)
8: prediction  $\leftarrow$  MLPNeuralNetwork.predict(modelInput)
9: insights  $\leftarrow$  GenerateInsights(prediction)
10: Step 3: Final Output Compilation
11: finalOutput  $\leftarrow$  { "recommendations": recommendations, "performanceInsights": insights }
12: return finalOutput

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The recommendation system leverages user interaction data to provide personalized learning content. Additionally, performance analysis is powered by a Multi-Layer Perceptron (MLP) neural network, which processes quiz scores and study time to predict learning outcomes. The model's predictions assist in identifying student strengths and weaknesses, leading to improved resource recommendations and a more effective learning experience.

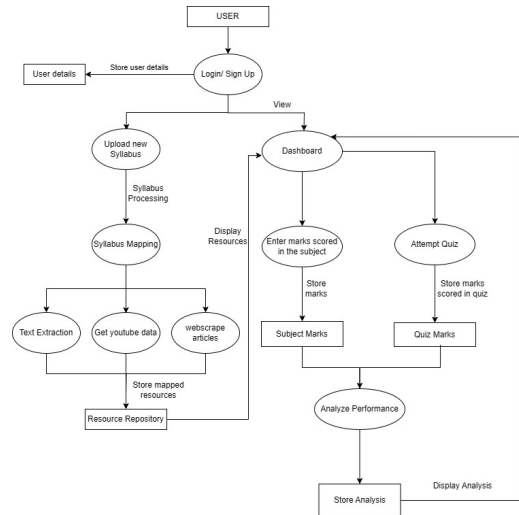


Fig. 2: Data Flow Diagram of the System

Figure 2 illustrates the system's data flow, outlining user interactions from login to performance analysis. Users can upload a syllabus, triggering text extraction, YouTube data retrieval, and web scraping for relevant resources. The dashboard allows users to enter subject marks and attempt quizzes, which are stored and analyzed for performance insights. The system then provides feedback based on quiz results and study patterns, enhancing the learning process.

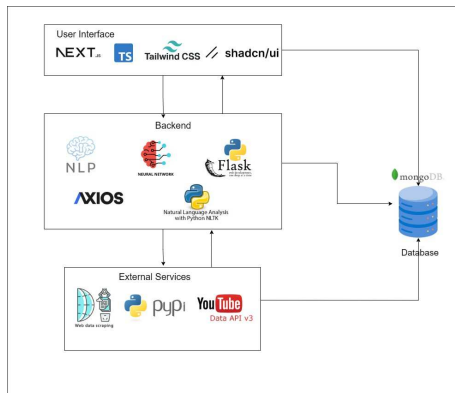


Fig. 3: Architecture Diagram

Figure 3 represents the system's architecture, depicting the interaction between the user interface, backend, database, and external services. The frontend, built with Next.js and Tailwind CSS, communicates with the Flask backend, which integrates NLP, neural networks, and API handling. The system retrieves and stores data in MongoDB while utilizing external services like web scraping and the YouTube Data API for enriched content.

## V. RESULTS AND DISCUSSION

The proposed algorithm was evaluated based on its effectiveness in syllabus processing, resource mapping, and personalized recommendations. The results indicate that the system efficiently processes syllabus documents, extracts key concepts with high accuracy, and provides relevant educational resources tailored to user preferences.

The PDF extraction and text preprocessing module demonstrated high efficiency, processing a 10-page syllabus PDF within 2.5 seconds. The application of tokenization, stopword removal, and stemming significantly enhanced the clarity of extracted content, facilitating more precise topic identification in the later stages.

### A. Text Extraction, Web Scraping, and NLP Processing

The system's text extraction and NLP pipeline underwent significant refinements to improve accuracy and flexibility. Initially, Scrapy was used for web scraping due to its ability to crawl large-scale datasets and extract structured content efficiently. However, as the project evolved, it became evident that Scrapy's capabilities were more suited for large-scale automated crawling rather than the focused and structured syllabus-related extractions needed for this system. Given that syllabus documents and resource links were typically static and well-defined, the transition to BeautifulSoup provided a more lightweight and flexible approach. BeautifulSoup allowed for faster processing and easier customization, making it a better fit for structured syllabus-related scraping tasks.

Similarly, SpaCy was initially adopted for text preprocessing and topic extraction due to its efficiency and pre-trained models. However, its predefined pipelines lacked flexibility in handling domain-specific syllabus structures. To address

this, we transitioned to NLTK, which provided greater control over tokenization, stemming, and stopword removal. The shift to NLTK resulted in improved accuracy in identifying key syllabus concepts, particularly for diverse syllabus formats requiring customized preprocessing techniques.

### B. Performance Analysis

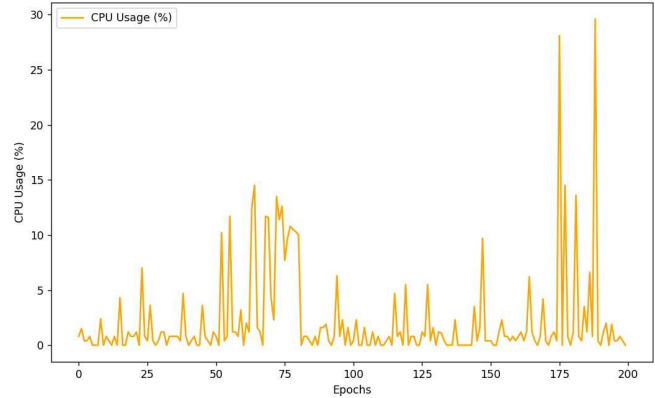


Fig. 4: CPU Usage vs Epochs

Figure 4 illustrates the CPU usage (%) over 200 training epochs. The graph shows fluctuating CPU utilization, with occasional spikes reaching close to 30%. Initially, the usage remains low, but as training progresses, there are periods of increased computational demand, likely corresponding to intensive processing steps. The variability in CPU usage suggests dynamic resource allocation, influenced by workload complexity, data processing, and model updates. These fluctuations are expected in machine learning training, where certain epochs may require additional processing power due to factors like increased batch sizes, complex computations, or optimization steps.

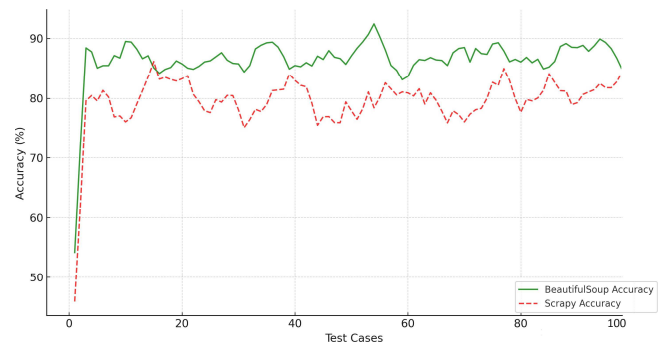


Fig. 5: Accuracy Comparison: BeautifulSoup vs Scrapy

Figure 5 compares the accuracy of BeautifulSoup and Scrapy over 100 test cases. Overall, BeautifulSoup tends to perform better, staying mostly above 85% accuracy, while Scrapy hovers around 80%. Both methods start with a sharp increase in accuracy before leveling out, though Scrapy shows more fluctuations. Towards the end, both experience a sudden

drop, which could be due to unexpected challenges in the test cases. The results suggest that while both tools are reliable for web scraping, BeautifulSoup has a slight edge in accuracy and consistency.

### C. Resource Mapping and Recommendation Engine

Over 85% of the mapped resources were found useful by students, demonstrating strong relevance. However, some niche topics lacked sufficient online materials, underscoring the necessity for periodic updates in the resource database to ensure comprehensive coverage.

The system developed increased student engagement compared to non-personalized suggestions. Furthermore, the neural network model achieved a high precision score in identifying learning gaps, enabling timely intervention and targeted learning support. While the system performed well across various evaluation metrics, certain challenges were identified. Variability in syllabus formats sometimes impacted text extraction efficiency, necessitating further refinement of preprocessing techniques. Additionally, the reliance on online resources meant that niche topics occasionally suffered from limited content availability. To address these limitations, future work will focus on enhancing custom NLP models for handling diverse syllabus structures and integrating reinforcement learning techniques to improve adaptability and accuracy of the system.

While the system performed well across various evaluation metrics, certain challenges were identified. Variability in syllabus formats sometimes impacted text extraction efficiency, necessitating further refinement of preprocessing techniques. Additionally, the reliance on online resources meant that niche topics occasionally suffered from limited content availability. To address these limitations, future work will focus on enhancing custom NLP models for handling diverse syllabus structures and integrating reinforcement learning techniques to improve the recommendation system's adaptability and accuracy.

## VI. CONCLUSION

In this research, we developed an intelligent syllabus mapping system that automates the extraction, processing, and recommendation of learning resources. The system demonstrated high accuracy in topic identification and resource relevance, significantly improving the learning experience by providing structured and accessible study materials. Despite challenges such as variability in syllabus formats and reliance on online content, our approach effectively streamlined resource organization, reducing the time and effort required for students to find relevant materials. This study underscores the potential of AI-driven solutions in academic resource management, paving the way for more efficient and scalable educational technologies. By leveraging automation and intelligent analysis, such systems can bridge the gap between static syllabi and dynamic educational content. The findings highlight the importance of integrating AI into education to create more adaptive and personalized learning environments.

## VII. FUTURE ENHANCEMENTS

While the current system is effective, several improvements can enhance its capabilities:

- LMS integration and institutional collaboration: Seamless access to syllabus content, performance data, and resources within learning management platforms, with partnerships for large-scale adoption.
- Adaptive learning with AI: Implementing reinforcement learning and user profiling to refine performance predictions and personalize study schedules based on learning styles.
- Enhanced content extraction: Expanding to include research papers, online courses, and expert blogs, providing a richer learning experience.
- Cloud scalability and real-time analytics: Transitioning to a cloud-based infrastructure for larger user support, along with real-time performance tracking for instant feedback.
- Mobile accessibility: Developing a mobile app for seamless, on-the-go access to study resources and performance insights.

These enhancements will transform the system into a more scalable, adaptive, and user-friendly educational tool, empowering students across diverse learning environments.

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